

LOSS-INFORMED TRANSFORMER TRAINING WITH LEARNED RATIONAL ACTIVATIONS

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ABSTRACT

Learning an activation changes how weight updates propagate through a network. We use this evolving response to improve Transformer pretraining through loss-informed coordination of matrix updates. GRAIN (Groupwise Rational Activation with Internal Normalization) learns rational response shapes in normalized feature coordinates and restores the group scale at its output. TILLER (Tangent-Informed Loss Ledger for Equal-budget Reweighting) uses these learned responses to orient matrix directions and measure their loss contributions at shared activation positions. A compact loss ledger retains a history of these responses, capturing how directions reinforce or offset one another at common positions. Combining this information with current gradients, TILLER jointly allocates update strength across groups and layers while preserving the combined direction norm of the feed-forward updates at each step. In language-model pretraining, GRAIN with TILLER achieves the lowest mean validation loss among the evaluated adaptive and matrix-based optimizer baselines across every tested corpus and model size at matched training-token budgets.

1 INTRODUCTION

Learning an activation function gives a Transformer control over both its feature representation and the response of those features to weight updates. Fixed nonlinearities such as GELU and the SiLU gate in SwiGLU provide effective transformations (Hendrycks & Gimpel, 2016; Shazeer, 2020). Rational activations make the response shape trainable (Molina et al., 2020), with approximation advantages that motivate their use as expressive neural representations (Boullé et al., 2020; Tang & Townsend, 2026). Group-shared rational functions bring this flexibility to Transformer feed-forward blocks (Yang & Wang, 2025). The changing response also supplies information for optimizing the matrices around it: learning the nonlinearity changes how an update propagates through the network.

The incoming and outgoing projections expose two complementary paths. An incoming update changes features through the activation Jacobian; an outgoing update recombines the activated features. Relating these changes to the downstream loss gives a functional description of each matrix direction. The interaction between directions is equally important. Two groups can have the same average descent signal while their loss responses reinforce or offset one another at shared activation positions. Separate descent scores leave these relationships unresolved. Measuring the responses together makes them available for choosing the strengths of simultaneous updates.

We introduce GRAIN, Groupwise Rational Activation with Internal Normalization, as the learned representation for this construction. Each feature group evaluates its own rational function on RMS-normalized coordinates and multiplies the result by the same normalization scale (Zhang & Sennrich, 2019). The learned coefficients control the response to relative feature magnitudes, while the output retains a dependence on input amplitude. Its Jacobian describes both the local rational slope and the effect of changing the group scale. The features and their derivatives provide the two response paths used to guide optimization.

Matrix-aware optimization uses matrix structure and activation sensitivity to shape updates (Martens & Grosse, 2015; Vyas et al., 2025; Liu et al., 2025). TILLER, Tangent-Informed Loss Ledger for Equal-budget Reweighting, uses the learned responses to construct matrix directions and coordinate their joint effect on the loss. It uses the response geometry to rotate an orthogonalized momentum direction along a factorized adaptive tangent, with orientation selected by the gradient (Shazeer &

Stern, 2018). It then records the selected feed-forward directions’ loss responses at shared positions in a compact temporal ledger. Together with the current gradients, this history defines a joint model for allocating update strength across groups and layers. The allocation preserves the combined norm of that step’s feed-forward directions, so the ledger changes where the update acts according to the responses it produces.

This construction gives two training methods. Full TILLER continually adapts directions and their joint allocation throughout training. The staged method applies TILLER for an initial interval, then continues from the learned model with a chosen optimizer. GRAIN remains trainable in both methods. Full TILLER maintains loss-response coordination along the complete trajectory; staged training combines a coordinated training stage with the update rule and computational cost of its continuation optimizer.

Across five pretraining corpora and both evaluated model sizes, GRAIN with full TILLER achieves the lowest mean validation loss among the tested activation–optimizer configurations at matched token budgets. The comparisons include AdamW, Muon, SOAP, and factorized adaptive optimizers. At the larger model size, the evaluated staged configuration also achieves lower mean validation loss than every baseline, while using less total training time than full TILLER on every corpus.

[TODO: Experiment section]

2 RELATED WORK

Padé Activation Units learn rational curves jointly with network weights (Molina et al., 2020). Kolmogorov–Arnold Transformer (KAT) implements group-shared rational functions followed by linear maps; its reported configuration shares numerator coefficients across groups (Yang & Wang, 2025). GRAIN evaluates each group’s rational function in RMS-normalized coordinates (Zhang & Sennrich, 2019) and restores the group scale at its output. It retains group sharing while learning both numerator and denominator coefficients independently for each group.

Approximation theory establishes parameter-efficiency separations between rational and ReLU networks (Boullé et al., 2020). *Rational Neural Networks have Expressivity Advantages*, by Maosen Tang and Alex Townsend, extends these comparisons to bounded-parameter smooth-activation networks and analyzes coefficient redundancy under learned affine normalization (Tang & Townsend, 2026). GRAIN uses an internal normalization–rescaling map with no additional learned affine transform.

Muon orthogonalizes momentum (Jordan et al., 2024), while Adafactor estimates gradient second moments from row and column statistics (Shazeer & Stern, 2018). TILLER combines these direction constructions using the learned activation response. Loss-weighted features and Jacobians determine the extent of a rotation from the polar direction along an adaptive tangent, with its orientation selected by the current gradient.

Joint update models differ in their search directions and the information used to compare them. K-FAC constructs parameter-space Fisher approximations from activation and backpropagated-derivative moments, including cross-layer blocks (Martens & Grosse, 2015). Newton–Muon derives input-second-moment preconditioning from a matrix quadratic surrogate (Du & Su, 2026). Krylov Subspace Descent optimizes the nonlinear objective over a curvature-generated Krylov subspace (Vinyals & Povey, 2012). TILLER places its quadratic model on coefficients multiplying selected pairs of incoming and outgoing group directions. Current gradients supply the linear term. A discounted history of shared-position response probes, augmented between probes by current-gradient information, supplies the quadratic penalty. Its cross terms couple coefficients across groups and layers; the allocation preserves the combined feed-forward direction norm before weight decay.

Optimizer switching combines update rules along a training trajectory. SWATS selects an Adam-to-SGD transition using a projection-based trigger (Keskar & Socher, 2017). Our staged method applies TILLER until a chosen step, then continues with a selected optimizer while retaining the learned parameters and trainable GRAIN activations.

3 METHOD

GRAIN learns the feed-forward activation response; TILLER uses this response to construct and coordinate matrix updates. We develop their shared construction and its full-horizon and staged training procedures.

3.1 GRAIN: GROUPWISE RATIONAL ACTIVATIONS

GRAIN (Groupwise Rational Activation with Internal Normalization) learns an activation in group-normalized coordinates and carries the group scale into its output. Each feature group in a Transformer feed-forward block has its own rational curve, following groupwise parameter sharing (Yang & Wang, 2025).

For a token representation $x \in \mathbb{R}^d$, partition the intermediate features into G groups of width w . Group g has incoming and outgoing projections $A_g \in \mathbb{R}^{w \times d}$ and $B_g \in \mathbb{R}^{d \times w}$. Writing $z_g = A_g x$, the block computes

$$\rho_g = \sqrt{\|z_g\|_2^2/w + \epsilon_{\text{act}}}, \quad h_g = \rho_g \phi_g(z_g/\rho_g), \quad y = \sum_{g=1}^G B_g h_g. \quad (1)$$

Here ρ_g is the root mean square for this token and group, stabilized by $\epsilon_{\text{act}} > 0$. Normalization makes relative feature values the input to the learned curve; multiplication by the same ρ_g reintroduces the group’s scale. For a linear response $\phi_g(u) = au$, these operations cancel exactly, giving $h_g = az_g$.

The scalar response uses learned rational coefficients (Molina et al., 2020) and is applied elementwise:

$$\phi_g(u) = \frac{\sum_{j=0}^5 a_{gj} u^j}{1 + \sum_{j=1}^4 |b_{gj}| |u|^j}. \quad (2)$$

Its denominator is at least one, excluding real poles. Each group learns its own numerator and denominator coefficients, shared across its features and token positions. The functions ϕ_g start from a common SiLU approximation and are trained jointly with the projection weights.

The RMS depends on all features in a group, so an incoming update can couple their responses through the full group Jacobian $J_g = \partial h_g / \partial z_g$. An outgoing update recombines the current features h_g . TILLER uses the Jacobian and features to orient matrix updates and evaluate their first-order effects on the loss. Appendix A derives the Jacobian and its action on a direction.

3.2 TILLER: LOSS-INFORMED UPDATE COORDINATION

TILLER uses the learned activation response to orient matrix directions and choose their strengths jointly across groups and layers.

3.2.1 RESPONSE-GUIDED MATRIX DIRECTIONS

For L layers, index the $C = LG$ layer–group pairs by $i = (\ell, g)$, with incoming and outgoing roles $r \in \{A, B\}$. For each matrix, TILLER forms a polar direction p_i^r and a factorized adaptive direction a_i^r from the same momentum (Jordan et al., 2024; Liu et al., 2025; Shazeer & Stern, 2018). The polar map rebalances singular values; factorized adaptation rescales entries using row and column gradient second moments. Removing the adaptive direction’s component parallel to the polar direction gives a tangent along which to rotate the update.

Two loss-weighted response statistics in $[0, 1]$ control the rotation. Normalized participation π_i^r measures the breadth of response energy: across Jacobian modes for the incoming role and feature coordinates for the outgoing role. Congruence κ_i^r is a weighted Frobenius cosine between the current response kernels, $J_i J_i^\top$ or $h_i h_i^\top$, and those evaluated with the initial rational coefficients at the same current inputs. Suppressing the role index, the rotation candidate is

$$\begin{aligned} a_i^\perp &= a_i - \frac{\langle a_i, p_i \rangle}{\|p_i\|_F^2} p_i, & e_i &= \pi_i (1 - \kappa_i^2), \\ \tilde{q}_i &= \sqrt{1 - e_i} p_i + \sigma_i \sqrt{e_i} \frac{\|p_i\|_F}{\|a_i^\perp\|_F} a_i^\perp. \end{aligned} \quad (3)$$

Here $\sigma_i \in \{-1, 1\}$ makes the tangent’s contraction with the current gradient nonnegative. Orthogonality preserves the parent norm. An unchanged response gives $e_i = 0$; a broad response with low congruence permits a larger rotation. A finite, nondegenerate candidate is selected when both it and its parent have positive gradient contraction; otherwise the parent is retained. Positive norm calibration gives the direction q_i used below. Appendix B specifies these operations.

3.2.2 JOINT LOSS-RESPONSE ALLOCATION

One signed coefficient c_i scales both directions (q_i^A, q_i^B) of each group, preserving the relative calibration of its two paths. To measure interactions among groups, sample K example–token positions shared across layers. At position k , let $x_{k\ell}$ be the block input and $\xi_{k\ell}$ the derivative of the summed minibatch loss with respect to the block output, scaled by the gradient-clipping factor. The response matrix $S_t \in \mathbb{R}^{K \times C}$ contains

$$S_{t,ki} = \langle \xi_{k\ell}, B_i J_{ki} q_i^A x_{k\ell} + q_i^B h_{ki} \rangle. \quad (4)$$

Each row collects the groups’ contributions through one activation position to the downstream loss derivative. For coefficients c , $S_t c$ combines these contributions. The cross terms in $\|S_t c\|_2^2/K$ distinguish reinforcement and cancellation that separate mean descent scores omit.

The loss ledger retains a discounted estimate G_t of the response second moment $S_t^\top S_t/K$ using a Robust Frequent Directions sketch (Ghashami et al., 2016; Luo et al., 2019). Its columns keep the same layer–group identities as directions evolve. Probes refreshed every eight steps measure positional variation; synthetic rows aligned with current gradient contractions update the history between probes, retaining the latest measured response scale. Together with the current descent signal $s_{t,i} = \langle g_i^A, q_i^A \rangle + \langle g_i^B, q_i^B \rangle$, this history defines the coefficient model

$$F_t(c) = -\eta s_t^\top c + \frac{\eta^2}{2} (c^\top G_t c + 2r_t^\top c), \quad \sum_i \omega_i c_i^2 = \sum_i \omega_i, \quad \omega_i = \|q_i^A\|_F^2 + \|q_i^B\|_F^2. \quad (5)$$

Here g_i^A, g_i^B are clipped gradients, η is the learning rate, and r_t records the response cross term with weight decay at rate λ_{wd} . The linear term rewards current descent. The quadratic term penalizes historical squared responses of the combined directions and weight decay. Its cross terms make the allocation sensitive to where recorded directions reinforce or cancel one another. The constraint preserves the combined norm of the paired feed-forward directions before weight decay. A projected solve proposes coefficients; acceptance requires a lower F_t than the all-ones allocation, the norm constraint, and positive layerwise gradient and momentum contractions for both allocations. Otherwise the all-ones allocation is retained. Appendix C gives the ledger and solver.

The selected coefficients update each incoming projection as $A_i \leftarrow (1 - \eta\lambda_{\text{wd}})A_i - \eta c_i q_i^A$, and analogously for B_i . Attention uses response-guided rotations with layerwise GRAIN summaries. Rational coefficients and the remaining parameter classes use AdamW (Loshchilov & Hutter, 2019).

Full TILLER applies these updates throughout the training horizon T . Its directions and joint allocation combine periodically refreshed GRAIN responses with current gradient information at every step, maintaining loss-response coordination along the full trajectory.

Staged training applies the same updates through a chosen step $T_s < T$, then continues the feed-forward and attention matrices with a selected optimizer. It retains the learned model and global learning-rate schedule, with an optimizer-specific state transition. GRAIN remains trainable, and its AdamW state continues across the transition. This method combines a coordinated training stage with the update rule and computational cost of its continuation optimizer. Table 1 presents both procedures.

4 EXPERIMENTS

[TODO: Experiment section]

5 DISCUSSION

6 CONCLUSION

AI USE STATEMENT

Generative AI tools assisted with method development, experimental planning, implementation, mathematical derivations and proof checking, literature analysis, result interpretation, and manuscript drafting and revision.

REFERENCES

- Nicolas Boullé, Yuji Nakatsukasa, and Alex Townsend. Rational neural networks. In *Advances in Neural Information Processing Systems*, volume 33, pp. 14243–14253, 2020. URL <https://proceedings.neurips.cc/paper/2020/hash/a3f390d88e4c41f2747bfa2f1b5f87db-Abstract.html>.
- Zhehang Du and Weijie Su. The Newton–Muon optimizer, 2026. URL <https://arxiv.org/abs/2604.01472>.
- Mina Ghashami, Edo Liberty, Jeff M. Phillips, and David P. Woodruff. Frequent directions: Simple and deterministic matrix sketching. *SIAM Journal on Computing*, 45(5):1762–1792, 2016. doi: 10.1137/15M1009718.
- Dan Hendrycks and Kevin Gimpel. Gaussian error linear units (GELUs), 2016. URL <https://arxiv.org/abs/1606.08415>.
- Keller Jordan, Yuchen Jin, Vlado Boza, Jiacheng You, Franz Cesista, Laker Newhouse, and Jeremy Bernstein. Muon: An optimizer for hidden layers in neural networks, 2024. URL <https://kellerjordan.github.io/posts/muon/>.
- Nitish Shirish Keskar and Richard Socher. Improving generalization performance by switching from Adam to SGD, 2017. URL <https://arxiv.org/abs/1712.07628>.
- Cornelius Lanczos. An iteration method for the solution of the eigenvalue problem of linear differential and integral operators. *Journal of Research of the National Bureau of Standards*, 45(4): 255–282, 1950. doi: 10.6028/jres.045.026.
- Jingyuan Liu, Jianlin Su, Xingcheng Yao, Zhejun Jiang, Guokun Lai, Yulun Du, Yidao Qin, Weixin Xu, Enzhe Lu, Junjie Yan, Yanru Chen, Huabin Zheng, Yibo Liu, Shaowei Liu, Bohong Yin, Weiran He, Han Zhu, Yuzhi Wang, Jianzhou Wang, Mengnan Dong, Zheng Zhang, Yongsheng Kang, Hao Zhang, Xinran Xu, Yutao Zhang, Yuxin Wu, Xinyu Zhou, and Zhilin Yang. Muon is scalable for LLM training, 2025. URL <https://arxiv.org/abs/2502.16982>.
- Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019. URL <https://openreview.net/forum?id=Bkg6RiCqY7>.
- Luo Luo, Cheng Chen, Zhihua Zhang, Wu-Jun Li, and Tong Zhang. Robust frequent directions with application in online learning. *Journal of Machine Learning Research*, 20(45):1–41, 2019. URL <https://jmlr.org/papers/v20/17-773.html>.
- James Martens and Roger Grosse. Optimizing neural networks with Kronecker-factored approximate curvature. In *Proceedings of the 32nd International Conference on Machine Learning*, volume 37 of *Proceedings of Machine Learning Research*, pp. 2408–2417. PMLR, 2015. URL <https://proceedings.mlr.press/v37/martens15.html>.
- Alejandro Molina, Patrick Schramowski, and Kristian Kersting. Padé activation units: End-to-end learning of flexible activation functions in deep networks. In *International Conference on Learning Representations*, 2020. URL <https://openreview.net/forum?id=BJlBSkHtDS>.

- Noam Shazeer. GLU variants improve transformer, 2020. URL <https://arxiv.org/abs/2002.05202>.
- Noam Shazeer and Mitchell Stern. Adafactor: Adaptive learning rates with sublinear memory cost. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pp. 4596–4604. PMLR, 2018. URL <https://proceedings.mlr.press/v80/shazeer18a.html>.
- Maosen Tang and Alex Townsend. Rational neural networks have expressivity advantages, 2026. URL <https://arxiv.org/abs/2602.12390>.
- Oriol Vinyals and Daniel Povey. Krylov subspace descent for deep learning. In *Proceedings of the Fifteenth International Conference on Artificial Intelligence and Statistics*, volume 22 of *Proceedings of Machine Learning Research*, pp. 1261–1268. PMLR, 2012. URL <https://proceedings.mlr.press/v22/vinyals12.html>.
- Nikhil Vyas, Depen Morwani, Rosie Zhao, Itai Shapira, David Brandfonbrener, Lucas Janson, and Sham Kakade. SOAP: Improving and stabilizing Shampoo using Adam for language modeling. In *International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/e988664070e9591f93fdcf605f7dc623-Abstract-Conference.html.
- Xingyi Yang and Xinchao Wang. Kolmogorov–Arnold transformer. In *International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/hash/bd611df389a2e2fd6baa43c381b2b012-Abstract-Conference.html.
- Biao Zhang and Rico Sennrich. Root mean square layer normalization. In *Advances in Neural Information Processing Systems*, volume 32, 2019. URL <https://proceedings.neurips.cc/paper/2019/hash/1e8a19426224ca89e83cef47f1e7f53b-Abstract.html>.

A JACOBIAN OF THE NORMALIZED RATIONAL ACTIVATION

An incoming matrix update changes both the normalized arguments of GRAIN and the scale restored at its output. Differentiating both paths gives the response used by TILLER. For one group, suppress the layer and group indices and write $z = Ax \in \mathbb{R}^w$:

$$\Phi(z) = \rho\phi(u), \quad u = z/\rho, \quad \rho = (z^\top z/w + \epsilon_{\text{act}})^{1/2}. \quad (6)$$

The scalar rational function in equation 2 acts coordinatewise on u ; its coefficients are held fixed in the derivatives below.

The normalized coordinates describe relative feature magnitudes, while ρ carries the group amplitude. When $\epsilon_{\text{act}} = 0$, a nonzero input satisfies $\Phi(az) = a\Phi(z)$ for every $a > 0$: positive rescaling leaves the arguments of the learned curve unchanged and scales its output by a . The implemented positive stabilizer keeps the map defined at the origin. Retaining this stabilizer, the derivatives of the scale and normalized coordinates are

$$\nabla_z \rho = \frac{u}{w}, \quad \frac{\partial u}{\partial z} = \rho^{-1} \left(I - \frac{uu^\top}{w} \right). \quad (7)$$

When $\epsilon_{\text{act}} = 0$, the matrix in parentheses projects onto the subspace orthogonal to u . Normalization then removes a radial perturbation from the curve’s input, while the outer factor ρ retains its effect on output amplitude.

At differentiability points, let $f = \phi(u)$, $d_\phi = \phi'(u)$, and $r_\phi = f - u \odot d_\phi$. At a zero scalar argument, the absolute-value terms use the selected autodifferentiation value $\text{sign}(0) = 0$. At these kinks, subsequent uses of J denote the resulting selected linearization. The product rule combines the two response paths:

$$\begin{aligned} J = \frac{\partial \Phi}{\partial z} &= \frac{fu^\top}{w} + \text{Diag}(d_\phi) \left(I - \frac{uu^\top}{w} \right) \\ &= \text{Diag}(d_\phi) + \frac{r_\phi u^\top}{w}. \end{aligned} \quad (8)$$

The diagonal term applies the rational slope at each coordinate. The rank-one term couples the coordinates through their shared scale, with r_ϕ measuring the difference between the response value and its local linear part. For $\phi(u) = au$, this difference vanishes and $J = aI$, consistent with the exact cancellation $\Phi(z) = az$ even when $\epsilon_{\text{act}} > 0$. Learning a nonlinear curve changes both its coordinatewise slopes and this coupling.

The same decomposition separates changes in feature proportions from changes in amplitude. For a perturbation v with $u^\top v = 0$, the scale has zero first-order change and $Jv = d_\phi \odot v$. Along the input itself, $u^\top u/w = 1 - \epsilon_{\text{act}}/\rho^2$ gives $Jz = \Phi(z) - (\epsilon_{\text{act}}/\rho)r_\phi$. The radial response therefore recovers $Jz = \Phi(z)$ in the unstabilized case, as required by positive homogeneity.

TILLER applies this Jacobian through coordinatewise products and inner products. For an input perturbation $v \in \mathbb{R}^w$ and an output sensitivity $\zeta \in \mathbb{R}^w$,

$$Jv = d_\phi \odot v + r_\phi \frac{u^\top v}{w}, \quad J^\top \zeta = d_\phi \odot \zeta + u \frac{r_\phi^\top \zeta}{w}. \quad (9)$$

Each action requires one inner product after the response and slope have been evaluated, avoiding a dense $w \times w$ Jacobian. In the paired projection $B\Phi(Ax)$, an incoming matrix direction U acts through $BJUx$, whereas an outgoing direction V acts through $V\Phi(Ax)$. These two paths supply the Jacobian and feature summaries used in Appendix B.

B RESPONSE-GUIDED MATRIX DIRECTIONS

TILLER chooses the orientation of each matrix update before coordinating update strengths across groups. It forms two transformations of the same momentum: a groupwise Muon direction and a factorized adaptive direction. The adaptive component orthogonal to the Muon direction supplies a tangent. The learned response controls the rotation along this tangent, and the current gradient

determines its sign. Preserving the calibrated Muon norm leaves the joint allocation in Appendix C to distribute a fixed update budget.

For group $i = (\ell, g)$ and role $r \in \{A, B\}$, let $g_{t,i}^r$ denote the training gradient after accumulation, distributed averaging, and global clipping. Starting from $v_{0,i}^r = 0$, the first-moment buffer and Nesterov momentum are

$$v_{t,i}^r = \beta_1 v_{t-1,i}^r + (1 - \beta_1) g_{t,i}^r, \quad m_{t,i}^r = \beta_1 v_{t,i}^r + (1 - \beta_1) g_{t,i}^r. \quad (10)$$

Muon rebalances the singular values of this momentum with an approximate polar map $\mathcal{P}_5$ using five Newton–Schulz iterations (Jordan et al., 2024; Liu et al., 2025). The map acts on each $w \times d$ incoming row block and each $d \times w$ outgoing column block. For total rational intermediate width $H_r = Gw$, define the parent direction and calibration factors by

$$p_{t,i}^r = \chi_{\text{grp}} \mathcal{P}_5(m_{t,i}^r), \quad \chi_{\text{grp}} = \sqrt{\frac{\min\{H_r, d\}}{G \min\{w, d\}}}, \quad \mu_{\text{mlp}} = 0.2 \sqrt{\max\{H_r, d\}}. \quad (11)$$

An exact polar map with full block rank would give total squared Frobenius norm $G \min\{w, d\}$ across the group blocks. The factor χ_{grp} matches this nominal norm to $\min\{H_r, d\}$ for a whole projection, for either matrix role. The rotation is formed at the scale of $p_{t,i}^r$; the final Muon multiplier μ_{mlp} is applied after selection.

The adaptive transformation rescales momentum entries using the history of squared gradients (Shazeer & Stern, 2018). Suppress the group and role indices, initialize $R_0 = C_0 = 0$, and set

$$\begin{aligned} R_t &= \beta_2 R_{t-1} + (1 - \beta_2) \text{rowsum}(g_t^2), \\ C_t &= \beta_2 C_{t-1} + (1 - \beta_2) \text{colsum}(g_t^2), \\ V_{t,jk} &= \frac{R_{t,j} C_{t,k}}{\max\{\sum_s R_{t,s}, \delta_{\text{fp}}\} (1 - \beta_2^t)}, \quad a_t = \frac{m_t}{\sqrt{V_t} + \epsilon_{\text{opt}}}. \end{aligned} \quad (12)$$

Squares, square roots, and division are elementwise. Row and column sums commute with exponential averaging, so R_t and C_t are the corresponding marginals of the full second-moment history. Their normalized outer product approximates that history with additional storage linear in the matrix dimensions. Both p_t and a_t transform the same momentum: the former operates on singular values, while the latter adapts to the gradient scales of matrix entries. Numerical constants, including the denominator floor δ_{fp} , are specified in Appendix D.

The two directions define a rotation in the Frobenius geometry of matrix updates. Removing the component of a_t parallel to p_t leaves a tangent to the sphere through p_t . Moving along this tangent introduces entrywise gradient-scale information while preserving the parent’s norm. The response statistics below choose the rotation angle; they are computed separately for the incoming and outgoing paths.

The activation response determines how far the update rotates from the parent. At sampled position k , write J_{ki} for the group Jacobian and $h_{ki} = \Phi_i(A_i x_{k\ell})$ for its features. The incoming path acts through J_{ki} ; the outgoing path recombines h_{ki} . Their response breadth is measured by

$$\text{pr}_A(J) = \frac{[\text{tr}(JJ^\top)]^2}{w \|JJ^\top\|_F^2}, \quad \text{pr}_B(h) = \frac{\|h\|_2^4}{w \|h\|_4^4}. \quad (13)$$

The incoming ratio measures concentration across squared singular values; the outgoing ratio measures concentration across squared feature coordinates. Equal energy in m nonzero components gives participation m/w , while concentration in one gives $1/w$. Thus incoming breadth is spectral and outgoing breadth refers to the chosen feature coordinates.

Positions receive more weight when the corresponding path is more sensitive to the loss. Let $\xi_{k\ell} \in \mathbb{R}^d$ be the downstream derivative at the block output. With $\mathcal{R}_{ki}^A = J_{ki}$ and $\mathcal{R}_{ki}^B = h_{ki}$, define

$$\alpha_{ki}^A = \frac{\|J_{ki}^\top B_i^\top \xi_{k\ell}\|_2^2}{w}, \quad \alpha_{ki}^B = \frac{\|\xi_{k\ell}\|_2^2}{d}, \quad \pi_i^r = \frac{\sum_k \alpha_{ki}^r \text{pr}_r(\mathcal{R}_{ki}^r)}{\sum_k \alpha_{ki}^r}. \quad (14)$$

The incoming weight is the mean squared loss derivative with respect to the preactivation; the outgoing weight is the corresponding quantity at the block output, shared across groups in the layer.

These local sensitivities emphasize response geometry at positions where feature perturbations affect the loss. Their use separates the sensitivity weighting from the input or feature amplitude that also enters a full matrix-gradient norm.

We compare the current response with the initial rational curve at the same current inputs. For this comparison, define the kernels $\Gamma_{ki}^A = J_{ki} J_{ki}^\top$ and $\Gamma_{ki}^B = h_{ki} h_{ki}^\top$. The first describes how the Jacobian distributes an isotropic input perturbation over feature coordinates; the second records the feature vector’s orientation and magnitude. A superscript 0 denotes evaluation with the initial rational coefficients at the current normalized input and RMS scale. The loss-weighted kernel cosine, which we call congruence, is

$$\kappa_i^r = \frac{\sum_k \alpha_{ki}^r \langle \Gamma_{ki}^r, \Gamma_{ki}^{r,0} \rangle}{\sqrt{\left(\sum_k \alpha_{ki}^r \|\Gamma_{ki}^r\|_F^2 \right) \left(\sum_k \alpha_{ki}^r \|\Gamma_{ki}^{r,0}\|_F^2 \right)}}. \quad (15)$$

Here $\langle \cdot, \cdot \rangle$ is the Frobenius inner product. Congruence is the cosine between the two collections of kernels, with each position weighted by $\sqrt{\alpha_{ki}^r}$. Evaluating both curves at the current input isolates the effect of the learned coefficients on this geometry. The comparison is invariant to a common positive scaling of the kernel collection and to sign reversals of J or h ; it measures alignment of response energy. In particular, $\phi = c\phi^0$ with $c \neq 0$ scales both kernels by c^2 and gives $\kappa = 1$. Overall response gain still enters the joint loss scores through J and h , while this cosine controls rotation according to kernel alignment.

The polar and adaptive transformations use different information from the same momentum. The polar map rebalances singular values, whereas factorized adaptation incorporates row and column gradient scales. Learning the activation changes the response through which these directions act. We use the initial curve as a reference for controlling how much adaptive information enters the direction: kernel departure permits rotation away from the polar parent, while current response participation attenuates this rotation when response energy is concentrated. The squared tangent weight is $e_i^r = \pi_i^r (1 - (\kappa_i^r)^2)$. This rule relates the rotation angle to the angle between response kernels. If $\psi_i^r = \arccos \kappa_i^r$ is the kernel angle and ϑ_i^r is the direction rotation, then $\sin^2 \vartheta_i^r = \pi_i^r \sin^2 \psi_i^r$. Participation therefore attenuates the kernel angle: $\vartheta_i^r \leq \psi_i^r$, with equality at full participation. Aligned kernels retain the parent direction; a broad response whose kernel geometry has changed allows a larger adaptive component.

The statistics keep this rule within its angular range. For nonnegative component energies v_j , $\sum_j v_j^2 \leq (\sum_j v_j)^2 \leq w \sum_j v_j^2$ places each nonzero participation ratio in $[1/w, 1]$. Assigning zero to a zero response and taking the loss-weighted average gives $0 \leq \pi_i^r \leq 1$ when the total weight is positive. The kernel products satisfy $\langle J J^\top, J^0 J^{0\top} \rangle = \|J^\top J^0\|_F^2$ and $\langle h h^\top, h^0 h^{0\top} \rangle = (h^\top h^0)^2$. Nonnegativity and Cauchy–Schwarz give $0 \leq \kappa_i^r \leq 1$ whenever its denominator is positive. Together with the zero-response and zero-sensitivity conventions in Appendix D, these bounds give $0 \leq e_i^r \leq 1$.

To form the rotation, suppress time and role indices and remove the adaptive direction’s component parallel to the parent:

$$\begin{aligned} a_i^\perp &= a_i - \frac{\langle a_i, p_i \rangle}{\|p_i\|_F^2} p_i, \\ \tilde{q}_i &= \sqrt{1 - e_i} p_i + \sigma_i \sqrt{e_i} \frac{\|p_i\|_F}{\|a_i^\perp\|_F} a_i^\perp. \end{aligned} \quad (16)$$

For nonzero parent and tangent, this rotates p_i through $\vartheta_i \in [0, \pi/2]$. Orthogonality gives $\|\tilde{q}_i\|_F^2 = (1 - e_i) \|p_i\|_F^2 + e_i \|p_i\|_F^2 = \|p_i\|_F^2$ in exact arithmetic. We set $\sigma_i = 1$ when $\langle g_i, a_i^\perp \rangle \geq 0$ and $\sigma_i = -1$ otherwise. The tangent then contributes nonnegatively to $\langle g_i, \tilde{q}_i \rangle$, the first-order descent signal for a subtraction update.

Selection checks this descent signal conditional on the parent. An admissible candidate is used only when both $\langle g_i, p_i \rangle > 0$ and $\langle g_i, \tilde{q}_i \rangle > 0$; otherwise the parent is retained. Admissibility requires finite values and nondegenerate directions, with the exact arithmetic tests given in Appendix D. The

final direction q_i is μ_{mlp} times the selection, so $\|q_i\|_F = \|\mu_{\text{mlp}} p_i\|_F$ in exact arithmetic when the numerical floors are inactive. A fallback has this norm directly.

The attention projections help determine the representations processed by the learned feed-forward map in the same Transformer layer. We use that map’s response geometry as a shared layer signal for the attention rotation. Each attention matrix supplies its own momentum, factorized moments, tangent, and gradient-selected sign; only the rotation fraction is shared. Let $\bar{\pi}_\ell^r = G^{-1} \sum_g \pi_{\ell g}^r$, and form κ_ℓ^r by summing the numerator and the two squared-norm sums in equation 15 across groups before taking the cosine. The QKV and attention-output matrices share

$$e_\ell^{\text{att}} = \sqrt{\bar{\pi}_\ell^A \bar{\pi}_\ell^B} (1 - \kappa_\ell^A \kappa_\ell^B). \quad (17)$$

Both feed-forward paths contribute to the breadth factor, while departure in either kernel collection can increase the rotation. Each matrix uses equation 16 with this fraction. Appendix D specifies the attention polar blocks and calibration, and Appendix D specifies response refreshes. Attention uses the selected, calibrated direction directly. The joint coefficients derived next apply to the paired GRAIN projections.

C JOINT LOSS-RESPONSE COORDINATION

The directions in Appendix B specify how each matrix changes. Their strengths must also be chosen jointly: directions in disjoint parameter blocks can reinforce or offset one another through the network. TILLER measures their loss responses at shared activation positions, retains a discounted second moment of these responses, and uses it to allocate the current feed-forward direction norm. The gradient supplies the current descent signal; the response history supplies information about how the directions act together.

C.1 LOSS RESPONSES AT SHARED POSITIONS

For a layer–group pair $i = (\ell, g)$, fix the current direction (q_i^A, q_i^B) and a probe position k . Let $x_{k\ell}$ be the block input, $h_{ki} = \Phi_i(A_i x_{k\ell})$ its group features, and J_{ki} the Jacobian in equation 8. Differentiating $(B_i + \alpha q_i^B) \Phi_i((A_i + \alpha q_i^A) x_{k\ell})$ at $\alpha = 0$ gives the block-output perturbation and its contraction with the downstream loss derivative:

$$\begin{aligned} \delta y_{ki} &= B_i J_{ki} q_i^A x_{k\ell} + q_i^B h_{ki}, \\ S_{t,ki} &= \langle q_i^A x_{k\ell}, J_{ki}^\top B_i^\top \xi_{k\ell} \rangle + \langle q_i^B h_{ki}, \xi_{k\ell} \rangle. \end{aligned} \quad (18)$$

The incoming term changes the features through their Jacobian; the outgoing term changes their recombination. Here $\xi_{k\ell}$ is the derivative of the summed minibatch loss with respect to the block output at that position, multiplied by the realized global clipping factor. The cotangent collects that hidden position’s influence on the downstream loss terms. Thus each score records one position’s contribution to the loss derivative along a paired matrix direction.

The probe derivatives use the same loss scaling as the parameter gradients. Suppose each microbatch contains R positions and its mean loss is divided by the number M of accumulated microbatches before backpropagation. The captured derivative $\bar{\xi}_{k\ell}$ then includes the factor $1/(RM)$. We recover the summed-loss scale and apply the realized clipping factor χ_{clip} through

$$\xi_{k\ell} = RM \chi_{\text{clip}} \bar{\xi}_{k\ell}. \quad (19)$$

The clipping factor is held fixed when evaluating a direction. Averaging the scores over all positions recovers the corresponding clipped minibatch derivative. For the current paired directions, that derivative is available directly from the full parameter gradients:

$$s_{t,i} = \langle g_{t,i}^A, q_{t,i}^A \rangle + \langle g_{t,i}^B, q_{t,i}^B \rangle. \quad (20)$$

TILLER therefore uses s_t without probe approximation in the linear descent term. The matrix $S_t \in \mathbb{R}^{K \times C}$ records how these contributions vary across the K shared positions, where $C = LG$ is the number of layer–group pairs.

For an allocation c , the combined responses are $S_t c$. Their second moment includes the cross products between groups, so it can distinguish allocations that have the same individual descent scores. For

example, the columns $(2, 0)^\top$ and $(0, 2)^\top$ each have mean one and individual mean squared response two. Their sum has mean squared response four. Two identical columns $(2, 0)^\top$ have the same individual summaries, but their sum has mean squared response eight. At this fixed measurement, the cross terms identify response overlap that the individual summaries omit.

Weight decay enters the response model through the same calculation. Substitute $(\lambda_{\text{wd}} A_i, \lambda_{\text{wd}} B_i)$ for the paired direction in equation 18 and sum over all GRAIN groups to obtain $a_t^{\text{dec}} \in \mathbb{R}^K$. Expanding the squared response of the structured update and decay gives

$$\frac{1}{K} \|S_t c + a_t^{\text{dec}}\|_2^2 = c^\top \frac{S_t^\top S_t}{K} c + 2c^\top \frac{S_t^\top a_t^{\text{dec}}}{K} + \frac{\|a_t^{\text{dec}}\|_2^2}{K}. \quad (21)$$

The last term is independent of c . The first two give the response metric and its cross term with decay, which the ledger accumulates. We use this squared response to regularize the current linear descent model. The linear term values improvement along the current gradient; the penalty measures the joint magnitude of the proposed step in the recorded loss-response coordinates. At equal current descent, the model prefers the allocation with smaller recorded combined response, including decay. Cross terms make this comparison sensitive to reinforcement and cancellation, while the fixed direction norm prevents reducing the penalty simply by shrinking the entire structured update. The factors η and η^2 in equation 5 express linear descent and squared responses at the size of the applied step.

C.2 DISCOUNTED RESPONSE HISTORY

TILLER refreshes $K = 32$ probes at steps 1, 9, 17, ... Each probe refers to the same example-token position across layers, so a row of S_t combines responses that occurred together. A refresh recomputes participation, congruence, and scores at the current parameters; Appendix D.2 specifies the position selection. Between refreshes, the direction construction reuses the latest participation and congruence values, while momentum, adaptive moments, directions, and gradient contractions continue to change every step.

The intervening ledger updates use synthetic rows aligned with the current descent vector. Let $\nu = \|S_{\text{ref}}\|_{\text{F}}/\sqrt{K}$ be the RMS row norm of the latest measured response matrix. With d_t^{dec} denoting the current clipped-gradient contraction with the GRAIN decay direction, define

$$S_t = \mathbf{1}_K (\gamma_t s_t)^\top, \quad a_t^{\text{dec}} = \mathbf{1}_K \gamma_t d_t^{\text{dec}}, \quad (22)$$

$$\gamma_t = \begin{cases} \nu/\|s_t\|_2, & \|s_t\|_2 > 0, \\ 0, & \|s_t\|_2 = 0. \end{cases}$$

For $s_t \neq 0$, this second moment has rank at most one and trace $\|S_t\|_{\text{F}}^2/K = \nu^2$. The construction updates its orientation using the current gradient while retaining the latest measured response scale. Variation across positions is supplied by the probe refreshes.

The uncompressed target for this sequence of measured and synthetic rows is the discounted second moment

$$G_t^* = \beta_2 G_{t-1}^* + (1 - \beta_2) S_t^\top S_t / K, \quad G_1^* = S_1^\top S_1 / K. \quad (23)$$

We use $\beta_2 = 0.95$, the same decay as the factorized second moments. For any c , the quantity $c^\top G_t^* c$ is a discounted average of squared responses for that coefficient combination. Columns retain their layer-group identities, and each historical measurement retains the direction used when it was recorded. Discounting lets recently observed relationships between group update strengths guide the current allocation.

To store this history, TILLER retains a factor $\mathcal{B}_{t-1}$ and an isotropic remainder $\tau_{t-1} I$. The stored approximation is $\hat{G}_{t-1} = \mathcal{B}_{t-1}^\top \mathcal{B}_{t-1} / K + \tau_{t-1} I$. New rows enter the metric for the current coefficient selection at full resolution:

$$\mathcal{A}_t = \begin{bmatrix} \sqrt{\beta_2} \mathcal{B}_{t-1} \\ \sqrt{1 - \beta_2} S_t \end{bmatrix}, \quad G_t = \mathcal{A}_t^\top \mathcal{A}_t / K + \beta_2 \tau_{t-1} I, \quad (24)$$

$$r_t = \beta_2 r_{t-1} + (1 - \beta_2) S_t^\top a_t^{\text{dec}} / K.$$

At the first step, $\mathcal{A}_1 = \mathcal{B}_1 = S_1$, $\tau_0 = \tau_1 = 0$, and $r_1 = S_1^\top a_1^{\text{dec}} / K$. Thus G_t^* describes the uncompressed target, G_t is the metric used now, and $\hat{G}_t$ is the compressed metric saved for later steps. The current selection uses G_t before introducing the new compression error.

We compress $\mathcal{A}_t$ with Robust Frequent Directions (Ghashami et al., 2016; Luo et al., 2019). Let its singular values be $\sigma_1 \geq \sigma_2 \geq \dots$ and retain at most $r = 64$ rows. If its rank exceeds r , set $\Delta_t = \sigma_{r+1}^2$ and subtract this value from each retained squared singular value. With right singular vectors $V_{1:r}$, the stored representation becomes

$$\begin{aligned} \mathcal{B}_t &= \text{Diag}(\sqrt{\sigma_1^2 - \Delta_t}, \dots, \sqrt{\sigma_r^2 - \Delta_t}) V_{1:r}^\top, \\ \tau_t &= \beta_2 \tau_{t-1} + \frac{\Delta_t}{2K}. \end{aligned} \quad (25)$$

If the rank is at most r , we take $\Delta_t = 0$ and retain an exact factor. The retained rows describe the dominant response combinations. Adding half the shrinkage as an isotropic term balances the error between retained and discarded directions, as the following exact-arithmetic bound shows.

Proposition 1 (Response-metric approximation). *For $K > 0$, $\beta_2 \in [0, 1]$, and $\tau_{t-1} \geq 0$, let G_t and $(\mathcal{B}_t, \tau_t)$ be defined by equations 24 and 25. Then $\hat{G}_t = \mathcal{B}_t^\top \mathcal{B}_t / K + \tau_t I$ is positive semidefinite and*

$$\|\hat{G}_t - G_t\|_2 \leq \frac{\Delta_t}{2K}, \quad |c^\top (\hat{G}_t - G_t) c| \leq \frac{\Delta_t}{2K} \|c\|_2^2 \quad \text{for every } c \in \mathbb{R}^C. \quad (26)$$

Proof. Both terms of $\hat{G}_t$ are positive semidefinite. When $\Delta_t = 0$, the metrics coincide. Otherwise, use an orthonormal basis of right singular vectors of $\mathcal{A}_t$, completed by its null space. On each retained direction, shrinkage removes Δ_t/K and the isotropic term restores $\Delta_t/(2K)$, so the error is $-\Delta_t/(2K)$. On a discarded direction with squared singular value $s \in [0, \Delta_t]$, the error is $(\Delta_t/2 - s)/K$; on a null direction, it is $\Delta_t/(2K)$. All absolute eigenvalue errors are therefore at most $\Delta_t/(2K)$. The second inequality follows from $|c^\top E c| \leq \|E\|_2 \|c\|_2^2$ for symmetric E . $\square$

The quadratic-form bound controls how compression changes the response penalty for any allocation. Its effect on the history follows by writing $E_t = \hat{G}_t - G_t^*$. Each compression contributes $D_t = \hat{G}_t - G_t$ with $\|D_t\|_2 \leq \Delta_t/(2K)$, giving $E_t = \beta_2 E_{t-1} + D_t$ and $E_1 = 0$. Hence

$$\|E_t\|_2 \leq \sum_{j=2}^t \beta_2^{t-j} \frac{\Delta_j}{2K}. \quad (27)$$

Older compression errors decay along with older measurements. Because the fresh rows enter before compression, the current metric satisfies $G_t - G_t^* = \beta_2 E_{t-1}$. The bounds quantify the fidelity of both metrics to the discounted measured-and-synthetic response history, with current selection retaining the full fresh responses.

C.3 FIXED-NORM JOINT ALLOCATION

Coordination changes the strengths of the paired directions while preserving their combined norm at the current step. Define $\omega_i = \|q_i^A\|_F^2 + \|q_i^B\|_F^2$ and $\mathcal{E}_0 = \sum_i \omega_i$. Since the groups occupy disjoint parameter blocks, the structured feed-forward update has norm $\|\Delta\theta(c)\|_2^2 = \eta^2 \sum_i \omega_i c_i^2$. The constraint $\sum_i \omega_i c_i^2 = \mathcal{E}_0$ therefore matches the current all-ones allocation before weight decay. Under this constraint, the model in equation 5 balances current descent against recorded squared responses and their cross term with decay. The latter depend on the directions at the times they were recorded. Signed coefficients permit the model to assess their combined action across groups and layers.

For $\omega_i > 0$, set $W = \text{Diag}(\omega)$ and $y = W^{1/2}c$. Dividing F_t by η^2 gives

$$\begin{aligned} \mathcal{Q}_t(y) &= \frac{1}{2} y^\top M_t y - b_t^\top y, \quad \|y\|_2^2 = \mathcal{E}_0, \\ M_t &= W^{-1/2} G_t W^{-1/2}, \quad b_t = W^{-1/2} (s_t/\eta - r_t). \end{aligned} \quad (28)$$

This change of coordinates turns the weighted constraint into a sphere. Its stationarity equation is $(M_t + \lambda I)y = b_t$. Since M_t is positive semidefinite, choosing $\lambda > 0$ makes the system positive definite and the solution norm monotone in λ . TILLER uses this positive branch to construct a proposal. The solve requires finite inputs, positive group energies, a positive learning rate, and $b_t \neq 0$; Appendix D.3 specifies the numerical checks on this domain.

A reorthogonalized Lanczos construction starts from b_t and repeated metric actions (Lanczos, 1950). Orthogonal restarts can extend the basis when a block closes, giving Q_k with $k \leq \min\{32, C\}$ columns. The metric acts through its factors:

$$M_t v = W^{-1/2} \mathcal{A}_t^\top (\mathcal{A}_t W^{-1/2} v) / K + \beta_2 \tau_{t-1} W^{-1} v. \quad (29)$$

This evaluates the quadratic model without forming a dense $C \times C$ matrix. Let T_k^+ be the positive-semidefinite projection of the symmetric matrix accumulated by Lanczos. For $y = Q_k z$, diagonalize $T_k^+ = U \text{Diag}(\theta) U^\top$ and set $\gamma = U^\top Q_k^\top b_t$. The projected stationarity equation and norm constraint reduce to

$$z = U \text{Diag}((\theta + \lambda)^{-1}) \gamma, \quad \psi(\lambda) = \sum_j \frac{\gamma_j^2}{(\theta_j + \lambda)^2} = \mathcal{E}_0. \quad (30)$$

For nonzero γ and $\lambda > 0$, $\psi'(\lambda) = -2 \sum_j \gamma_j^2 / (\theta_j + \lambda)^3 < 0$, so there is at most one positive root. When the numerical bracket contains this root, bisection gives the proposal $c = W^{-1/2} Q_k z$. Otherwise, the method retains $c = \mathbf{1}$. Appendix D.3 specifies the bracket and termination settings.

The proposal is evaluated in the full C -dimensional model F_t . It is accepted only if it is finite, strictly improves on $F_t(\mathbf{1})$, and satisfies the prescribed norm tolerance. For every layer and each matrix role, acceptance also requires positive gradient and momentum contractions:

$$\sum_g c_{\ell g} \langle g_{\ell g}^r, q_{\ell g}^r \rangle > 0, \quad \sum_g c_{\ell g} \langle m_{\ell g}^r, q_{\ell g}^r \rangle > 0, \quad r \in \{A, B\}. \quad (31)$$

These contractions must be positive for both the proposal and $\mathbf{1}$. If an acceptance condition fails, the method retains the uncoordinated allocation. The improvement test uses F_t with the current metric G_t ; the projected model constructs the proposal.

D TRAINING ALGORITHMS AND NUMERICAL SPECIFICATION

Full-horizon and staged training share the same TILLER updates through a chosen step T_s . Setting $T_s = T$ applies these updates throughout the training horizon. For $T_s < T$, the procedure takes two additional inputs: a continuation update rule $\mathcal{U}$ for the feed-forward and attention matrices, and its state-initialization map $\mathcal{I}$. The map specifies the initial continuation state from the learned parameters, the TILLER state, and T_s , including moment buffers, internal step counts, and any required matrix grouping. This pair defines the staged member of the training family.

Table 1 gives the common procedure. The transition follows TILLER update T_s , so update $T_s + 1$ is the first application of $\mathcal{U}$. The model parameters and global learning-rate schedule continue across this boundary. AdamW moments and step counts also continue for the rational coefficients and other tensors assigned to AdamW. Thus GRAIN keeps learning after the matrix update rule changes, while the TILLER response measurements cease.

Alongside the matrix moment buffers (v, R, C) and AdamW state, TILLER stores the initial rational coefficients, the latest participation and congruence summaries, the measured RMS row norm ν , and the ledger state $(\mathcal{B}, \tau, r)$. Matrix moments are updated and directions and gradient contractions are formed every step. Thus cached response geometry guides current updates rather than reusing an earlier matrix direction.

D.1 ACTIVATION INITIALIZATION AND MATRIX DIRECTIONS

Every group begins with the same rational approximation to SiLU on $[-5, 5]$. In ascending powers, its coefficients are

$$(a_0, \dots, a_5) = (2.4915714620520875 \times 10^{-5}, 0.5000668663485823, 0.2499441908995771, 0.0526200910151016, 0.005525765640115002, 0.00024199321178747251), \quad (32)$$

$$(b_1, \dots, b_4) = (0.00022554017910995938, 0.10511420350691994, 2.8656992394109812 \times 10^{-5}, 0.0004816962110562025). \quad (33)$$

Table 1: Full-horizon and staged TILLER training with GRAIN. The continuation rule $\mathcal{U}$ and its state initializer $\mathcal{I}$ are inputs when $T_s < T$. Current response rows enter coefficient selection before compression, and rational coefficients are trained throughout.

Training procedure	
1	Given horizon T , switch point $T_s \in \{1, \dots, T\}$, schedule $\{\eta_t\}_{t=1}^T$, and, for $T_s < T$, the pair $(\mathcal{U}, \mathcal{I})$, with $\eta_t > 0$ during the TILLER stage, initialize GRAIN, zero matrix moments, AdamW state, and an empty ledger.
2	For $t = 1, \dots, T$:
3	Evaluate the loss, accumulate and average gradients, and apply global clipping. Capture aligned probes during the pass when $t \leq T_s$ and $t \in \{1, 9, 17, \dots\}$.
4	If $t \leq T_s$:
5	Update the matrix moments and form the parent and adaptive directions (equations 10–12).
6	Refresh participation and congruence if probes were captured. Use the latest summaries with the current directions to select the feed-forward and attention rotations (equations 16 and 17).
7	Compute s_t from current gradients and directions. On a refresh, rescale captured derivatives using the realized clipping factor and form S_t, a_t^{dec} ; otherwise form the synthetic rows in equation 22.
8	Form G_t, r_t , using the first-step initialization in Appendix C.2 when $t = 1$. Propose and select coefficients using Appendix C.3. Retain the compressed history in equation 25 for the next step.
9	Apply $A_i \leftarrow (1 - \eta_t \lambda_{\text{wd}})A_i - \eta_t c_i q_i^A$ and $B_i \leftarrow (1 - \eta_t \lambda_{\text{wd}})B_i - \eta_t c_i q_i^B$; apply the selected attention directions with weight decay.
10	Else:
11	Apply $\mathcal{U}$ to the feed-forward and attention matrices, with its current state and scheduled learning rate.
12	End if.
13	Update rational coefficients, embeddings, normalization parameters, and remaining tensors with AdamW.
14	If $t = T_s < T$, initialize the continuation state with $\mathcal{I}$ and stop collecting response measurements.
15	End for.

The ten coefficients subsequently learn independently in each group. The denominator in equation 2 remains at least one for finite coefficients. We use group width $w = 256$ and $\epsilon_{\text{act}} = 10^{-6}$. If H is the SwiGLU intermediate width, GRAIN uses $Gw = 3H/2$: its two projections contain $2d(3H/2) = 3dH$ dense weights, matching the three SwiGLU projections.

The polar direction is approximated by five Newton–Schulz iterations. For each matrix block, a tall input is transposed so that the Gram matrix uses the smaller dimension. The working matrix is cast to bfloat16 and divided by its Frobenius norm, accumulated in float32, lower bounded by 10^{-7} , and cast to bfloat16. With $(a, b, c) = (3.4445, -4.775, 2.0315)$, the iteration is

$$X_{j+1} = aX_j + b(X_j X_j^\top)X_j + c(X_j X_j^\top)^2 X_j. \quad (34)$$

The original orientation is restored after the fifth iteration. Factorized second moments are maintained in float32, with $\beta_1 = \beta_2 = 0.95$ and $\epsilon_{\text{opt}} = 10^{-8}$ as in Appendix B.

The feed-forward polar maps act on individual group blocks and use the calibration in equation 11. For attention, each of Q, K, and V is divided into four contiguous row blocks, and the output projection into four contiguous column blocks. Each block spans $d/4$ features. The stacked QKV parent is scaled by $1/\sqrt{3}$ to account for its three projections. Factorized moments and the tangent rotation use each whole attention matrix. After selection, a matrix of shape $n \times m$ receives the Muon scale $0.2\sqrt{\max\{n, m\}}$.

The rotation guard evaluates the geometry at the parent scale before the final Muon multiplier. It requires finite inputs, finite evaluated coefficients and candidate contraction, and strictly positive parent and adaptive squared norms. The tangent squared norm is computed as

$$d_\perp = \max \left\{ \|a_i\|_{\text{F}}^2 - \frac{\langle a_i, p_i \rangle^2}{\max\{\|p_i\|_{\text{F}}^2, \delta_{\text{fp}}\}}, 0 \right\}, \quad \delta_{\text{fp}} = \text{finfo}(\text{float32}).\text{tiny}. \quad (35)$$

The candidate requires $d_\perp > \delta_{\text{fp}}$; projection and tangent normalization use the same denominator floor. It is selected only when both its gradient contraction and the parent contraction are positive. For finite parent and adaptive directions, a failed guard selects the parent by setting its combination coefficient to one and the adaptive coefficient to zero. The resulting direction must pass a finite-value check before it is applied.

D.2 PROBE SELECTION AND RESPONSE CONVENTIONS

Global gradient clipping uses threshold one. If g_{raw} is the accumulated, distributed-averaged gradient before clipping, the probe cotangents use the same realized factor $\chi_{\text{clip}} = \min\{1, (\|g_{\text{raw}}\|_2 + 10^{-6})^{-1}\}$ in equation 19.

The $K = 32$ probes are selected deterministically from flattened example–token positions. With P distributed processes, process $p \in \{0, \dots, P-1\}$ contributes $K_p = \lfloor (p+1)K/P \rfloor - \lfloor pK/P \rfloor$ probes. For $M = 4$ accumulated microbatches, it initially captures $m_p = \lceil K_p/M \rceil$ positions from each microbatch. To select $m > 1$ positions from N rows, it uses zero-based indices $\lfloor j(N-1)/(m-1) \rfloor$ for $j = 0, \dots, m-1$; a single selected position uses index zero, and zero selected positions give an empty set. The same rule selects K_p rows from the concatenated Mm_p captured rows. Inputs, preactivations, features, and output derivatives use the same indices in every layer. Gathering these rows supplies the shared K -position response matrix.

Participation and congruence use conventions that keep their rotation fractions defined at zero responses. At an individual position, a zero Jacobian or feature vector has participation zero. After loss-sensitivity weighting within a group, if either matrix role has zero total sensitivity, both group participation values are set to one. For group congruence, if either role has a zero denominator, both values are set to one; equality of the rational coefficients to their initialization also sets both values to one. The congruence convention takes precedence in the product $\pi_i^r(1 - (\kappa_i^r)^2)$: an undefined comparison then gives zero departure and retains the parent orientation. Attention forms its congruence after summing the kernel statistics across groups and uses the same unit-congruence convention if either aggregate denominator vanishes. For the synthetic rows, a score norm no larger than $\text{finfo}(\text{float64})$. tiny is treated as zero in equation 22.

D.3 COEFFICIENT-SOLVER NUMERICS

Response factors and the persistent ledger are stored in float32; the coefficient solve and its acceptance tests use double precision. The implementation requires $C \geq 2$, finite response and contraction inputs, nonnegative isotropic residuals, $\omega_i > 0$ for every group, $\eta > 0$, and $b_t \neq 0$. Invalid inputs, including zero group energies or a zero right-hand side, raise an error. The all-ones fallback in Appendix C.3 applies after these input checks and a valid basis construction.

Lanczos uses $k = \min\{32, C\}$ basis vectors and two reorthogonalization passes against the accumulated basis at each step. Let v_j be its j th basis vector. A residual norm no greater than $64\epsilon_{\text{mach}} \max\{\|M_t v_j\|_2, 1\}$ triggers an attempted restart. The seed has coordinates $[u_j]_a = \cos(aj)$ for $a = 1, \dots, C$ and j starting at one. After two reorthogonalization passes, a seed with norm above the same threshold is normalized and starts a new block; its coupling to the preceding block is set to zero. If the seed is unusable, the routine uses the residual divided by its norm floored at $\text{finfo}(\text{float64})$. tiny, then checks the resulting basis. It requires finite basis and projected entries and $\|Q_k^\top Q_k - I\|_F \leq 2 \times 10^{-10}k$; failure raises an error.

The accumulated symmetric Lanczos matrix is eigendecomposed and its negative eigenvalues are clipped to zero. This enforces the positive-semidefinite projected model used by the positive-multiplier solve. With double-precision machine epsilon ϵ_{mach} , the multiplier bracket is

$$\lambda_{\text{lo}} = \epsilon_{\text{mach}}C, \quad \lambda_{\text{hi}} = \|b_t\|_2 / \sqrt{\mathcal{E}_0} + \lambda_{\text{max}}(T_k^+) + \lambda_{\text{lo}}. \quad (36)$$

Nonnegative projected eigenvalues and orthonormality of Q_k give $\psi(\lambda_{\text{hi}}) \leq \|b_t\|_2^2 / \lambda_{\text{hi}}^2 \leq \mathcal{E}_0$. If $\psi(\lambda_{\text{lo}}) \geq \mathcal{E}_0$, 64 bisection iterations give a proposal; otherwise the unit allocation is retained. The norm check is

$$\frac{|c^\top Wc - \mathcal{E}_0|}{\max\{\mathcal{E}_0, 1\}} \leq 10^{-8}. \quad (37)$$

Acceptance uses the original factor representation of G_t and the current gradient and momentum contractions. The norm tolerance applies to the double-precision proposal. Selected coefficients are then cast to float32 and multiply the float32 matrix directions; the resulting updates are cast to the parameter dtype.